

HELIA TAJABADI

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EDUCATION

Sharif University of Technology

Bachelor of Science in Electrical Engineering

Cumulative GPA: 18.58 / 20.00 | Major GPA: 19.11 / 20.00

Tehran, Iran

Expected 2028

Salam High School

Advanced studies in Number Theory, Geometry, Combinatorics, Algebra, and Graph Theory

Total GPA: 19.92 / 20.00

Tehran, Iran

2021–2024

PROJECTS

Fourier Optics *Course Project*

- Analyzed optical diffraction using a signal-processing framework, covering 2D Fourier transforms, Fourier-domain representations, Fresnel and Fraunhofer approximations, and spatial sampling.
- Modeled free-space propagation as a linear shift-invariant system and investigated the relationship between aperture functions and resulting diffraction intensity patterns.

Hidden Markov Anomaly Detection *Course Project*

- Implemented a sequence-aware anomaly detection framework based on Hidden Markov Models (HMMs), combining Viterbi decoding, a joint hidden-state/observation feature map, and One-Class SVM.

Tanaka's Central Limit Theorem *Course Project*

- Studied the Tanaka Central Limit Theorem and developed a proof based on convergence in the Wasserstein-2 metric.
- Investigated the relationship between weak convergence and Wasserstein convergence, including the role of second-moment convergence.

TEACHING EXPERIENCE

Sharif University of Technology — Teaching Assistant

Tehran, Iran

Probability & Statistics (Dr. Mojahedian) — Course Project Team

Fall 2026

Signal & Systems (Dr. Dastgheib) — Course Project Team

Fall 2026

Electrical Circuits (Dr. Fatemizadeh) — Designing Computational Exercises

Spring 2026

Electrical Circuits (Dr. Zarghani) — Designing Computational Exercises

Spring 2026

Electrical & Electronic Circuits (Dr. Koohi) — Exercise & Quiz Design; Final Exam Grading

Spring 2026

Mathematical Methods (Dr. PilAram) — Designing Theoretical and Computational Exercises

Spring

2026–Fall 2026

Fundamentals of Programming (Dr. Vosoughi Vahdat & Dr. Arasteh) — Designing Assignments

Fall 2025

Young Scholars' Club (YSC) — Instructor

Tehran, Iran

Mathematics Olympiad Instructor — Summer programs and independent school instruction

2024–2026

INTERNSHIP EXPERIENCE

Embedded Systems Intern *Smart Home Startup* — Microcontroller firmware development; ESP32 and ESP8266

School of Electrical & Computer Engineering, SBU, 2025

ACHIEVEMENTS

- Bronze Medal** — Iranian National Mathematics Olympiad
- Gold Medal** — Náboj Mathematics Competition
- Third Prize** — European mathematical cup
- Honorable Mention** — Combinatorics & Geometry Competition, Stanford SMT
- Top 11% Academic Ranking** — 23/208

SELECTED ADVANCED COURSEWORK AND INDEPENDENT STUDY

- Machine Learning** — Formal university course (19.8/20)

VOLUNTEER & LEADERSHIP EXPERIENCE

Teaching Assistant — YSC Summer Camp — Educational activities and preparation of educational materials and learning packages 2024–2026

Vice President — Electrical Engineering Scientific & Cultural Association (RESANA, Sharif University of Technology) — Academic, cultural, and student activities 2025–Present

CERTIFICATIONS & ADDITIONAL TRAINING

Fundamentals of MATLAB Programming (Quera College) — Completed with a Perfect Score July 2025

Head Start in MATLAB (Electrical Engineering Association, SBU) — 24-hour course; completed with an A++ grade January 2025

IoT Bootcamp (IEEE Student Branch, SBU) — Full attendance and strong performance 2025

TECHNICAL SKILLS

Programming C++, MATLAB, Python, Verilog

Tools MATLAB, LTspice, Vivado, Altium, Arduino, L^AT_EX

Languages Persian (Native), English (Professional Working Proficiency)